

Data and response format for polarization analysis using PolPy

Sujay Mate, Hancheng Li, Utkarsh Pathak, Yashowardhan Rai,
Nicolas De Angelis, Varun Bhalerao, Merlin Kole

1 Introduction

This document defines the standard data and response file format for polarization analysis using PolPy. Both formats aim to be compatible with the HEASARC standards defined in OGIP Calibration Memo [CAL/GEN/92-002](#) and with the [3ML](#) data analysis framework.

2 Polarization data file format

Generally, polarimeters use scattering polarimetry techniques where there is one active scattering element and one active absorbing element. Hence, to construct the azimuthal scattering histogram, a pair of events (“double events”) satisfying scattering and absorption criteria is necessary. As different instruments have different criteria to select such pairs, the standard data format should have a reduced format where the final scattering bin of each event pair is reported. Therefore, the data file should contain a single FITS extension named ‘POLEVENTS’ and the filename should have extension `.pevt`. The FITS extension should have the following format:

2.1 Extension header

The extension header should contain the following standard keywords defined in the OGIP format:

- `EXTNAME` (= ‘POLEVENTS’) - the name (*ie* type) of the extension.
- `TELESCOP` - the “telescope” (*ie* mission/satellite name).
- `INSTRUME` - the instrument/detector.
- `EMIN` - The minimum detected energy used to generate the data. The preferred units are keV.
- `EMAX` - The maximum detected energy used to generate the data. The preferred units are keV.
- `NCHANS` - Total number of PHA channels.
- `NSABINS` - Total number of output scattering angle bins.

In addition to these, because the polarization angle is measured with respect to the local North in the celestial frame, additional attitude information is needed to transform the data and templates correctly. The attitude information should be given as the following additional headers:

- **RAX** - Right Ascension of the instrument X-axis at the time of GRB. The preferred units are deg.
- **DECX** - Declination of the instrument X-axis at the time of GRB. The preferred units are deg.
- **RAZ** - Right Ascension of the instrument Z-axis at the time of GRB. The preferred units are deg.
- **DECZ** - Declination of the instrument Z-axis at the time of GRB. The preferred units are deg.
- **RAGRB** - Right Ascension of the source position. The preferred units are deg.
- **DECGRB** - Declination of the source position. The preferred units are deg.

2.1.1 Extension data

The extension data should be a BINTABLE with the following columns and data types:

1. *Time*, an 8-byte REAL scalar for each row containing the time of the event.
The FITS column name is **TIME**.
The recommended units are s.
2. *Chan*, a 2-byte or 4-byte INTEGER scalar giving the raw channel number for each row.
The FITS column name is **CHANNEL**.
(unitless).
3. *SABin*, a 2-byte INTEGER scalar giving the scattering angle bin for each row. The binning should be done with the highest possible resolution.
The FITS column name is **SABIN**
(unitless)
4. *Dead Time Fraction*, a 4-byte REAL scalar for each row containing the dead time fraction for the event.
The FITS column name is **DEADFRAC**.

3 Polarization response file format

A typical response needed for a polarization study requires the observed Scattering Angle (SA) distribution over a grid of Polarization Angle (PA) and Polarization Fraction (PF). This response has to be generated for a source spectrum individually. Hence, the raw response file should have the observed Scattering Angle distribution for 100% polarized photons with different input PAs simulated at different input energies (E), i.e. a matrix with shape $nE \times nPA \times nSA$. Along with this, another matrix with fully unpolarized photon simulations is needed to generate the response in the PF space. Therefore the response file should contain the following FITS extensions. The preferred filename extension for the file is **.prsp**.

- An extension containing input energy bounds (**INEBOUNDS**)
- An extension containing input polarization angle (**INPAVALS**)
- An extension containing the angle bounds on scattering angle (**SABOUNDS**)

- An extension containing the $nE \times nPA \times nSA$ matrix with 100% polarised photon simulations (SPECRESP POLMATRIX)
- An extension containing the $nE \times nSA$ matrix for unpolarized photon simulations (SPECRESP UNPOLMATRIX)

3.1 The INEBOUNDS extension

The INEBOUNDS extension lists the energy bounds on the input energy. The headers and data format for this extension are given below.

3.1.1 Extension header

The extension header should contain the following standard keywords defined in the OGIP format:

- EXTNAME (= 'INEBOUNDS') - the name (*ie* type) of the extension.
- TELESCOP - the “telescope” (*ie* mission/satellite name).
- INSTRUME - the instrument/detector.

In addition to this, it should have the following headers:

- EBINS - Total number of input energy bins.

3.1.2 Extension data

The extension data should be a BINTABLE with the following columns and data types:

1. E_{low} , a 4-byte REAL scalar for each row containing the lower energy bound of the input energy bin.
The FITS column name is `ENERG_LO`.
The recommended units are keV.
2. E_{high} , a 4-byte REAL scalar for each row containing the upper energy bound of the input energy bin.
The FITS column name is `ENERG_HI`.
The recommended units are keV.

3.2 The INPAVALS extension

The INPAVALS extension lists the input polarization angle values for 100% polarized photons. Note that it is the responsibility of the individual mission to define a local reference frame for the polarization angles and give that to 3ML for analysis. The header and data format for this extension are given below.

3.2.1 Extension header

The extension header should contain the following standard keywords defined in the OGIP format:

- EXTNAME (= 'INPAVALS') - the name (*ie* type) of the extension.
- TELESCOP - the “telescope” (*ie* mission/satellite name).

- INSTRUME - the instrument/detector.

In addition to this, it should have the following headers:

- PABINS - Total number of input polarization angle bins.

3.2.2 Extension data

The extension data should be a BINTABLE with the following columns and data types:

1. PA_{in} , a 4-byte REAL scalar for each row containing the input polarization angle value.
The FITS column name is PA_IN.
The recommended units are deg.

3.3 The SABOUNDS extension

The SABOUNDS extension lists the scattering angle bounds on the output scattering angle bin. The headers and data format for this extension are given below.

3.3.1 Extension header

The extension header should contain the following standard keywords defined in the OGIP format:

- EXTNAME (= 'SABOUNDS') - the name (*ie* type) of the extension.
- TELESCOP - the “telescope” (*ie* mission/satellite name).
- INSTRUME - the instrument/detector.

In addition to this, it should have the following headers:

- SABINS - Total number of output scattering angle bins.

3.3.2 Extension data

The extension data should be a BINTABLE with the following columns and data types:

1. $SABin$, a 2-byte INTEGER scalar representing each scattering bin value. The binning should represent the highest possible resolution.
The FITS column name is SABIN.
(unitless)
2. SA_{min} , a 4-byte REAL scalar for each row containing the lower angle bound of the scattering angle bin.
The FITS column name is SA_MIN.
The recommended units are deg.
3. SA_{max} , a 4-byte REAL scalar for each row containing the upper angle bound of the scattering angle bin.
The FITS column name is SA_MAX.
The recommended units are deg.

3.4 The SPECRESP POLMATRIX extension

The SPECRESP POLMATRIX extension stores the polarization response for the polarization angle and energy values given in the above two extensions. The header and data format for this extension are given below.

3.4.1 Extension header

The extension header should contain the following standard keywords defined in the OGIP format:

- **EXTNAME** (= ‘SPECRESP POLMATRIX’) - the name (*ie* type) of the extension. The SPECRESP is important as this specifies that the values in the matrix are normalised to the effective area in units of cm^2 .
- **TELESCOP** - the “telescope” (*ie* mission/satellite name).
- **INSTRUME** - the instrument/detector.

In addition to this, it should have the following headers:

- **EMIN** - The minimum detected energy used to generate the matrix. The preferred units are keV.
- **EMAX** - The maximum detected energy used to generate the matrix. The preferred units are keV.

3.4.2 Extension data

The extension data should be an IMAGE with dimensions $nE \times nPA \times nSA$. The data should be a 4-byte REAL scalar. The recommended units are cm^2 .

3.5 The SPECRESP UNPOLMATRIX extension

The SPECRESP UNPOLMATRIX extension stores the polarization response for the polarization angle and energy values given in the above two extensions. The header and data format for this extension are given below.

3.5.1 Extension header

The extension header should contain the following standard keywords defined in the OGIP format:

- **EXTNAME** (= ‘SPECRESP UNPOLMATRIX’) - the name (*ie* type) of the extension. The SPECRESP is important as this specifies that the values in the matrix are normalised to the effective area in units of cm^2 .
- **TELESCOP** - the “telescope” (*ie* mission/satellite name).
- **INSTRUME** - the instrument/detector.

In addition to this, it should have the following headers:

- **EMIN** - The minimum detected energy used to generate the matrix. The preferred units are keV.
- **EMAX** - The maximum detected energy used to generate the matrix. The preferred units are keV.

3.5.2 Extension data

The extension data should be an IMAGE with dimensions $nE \times nSA$. The data should be a 4-byte REAL scalar. The recommended units are cm^2 .